

9.2.2 Type II Error and the Choice of the Sample Size

As mentioned before, the size of α is chosen by ourselves. The size of β depends on the size of α and the sample size n .

How large should n be chosen such that $\beta = P(H_0 \mid H_1)$ is less than or equal to a limit value β_0 ?

Suppose we test $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$, while H_1 is true, i.e. $\mu = \mu_0 + \delta$, with $\delta > 0$.

$$\text{Check that: } C_R = \text{invNorm}(1 - \frac{1}{2}\alpha, \mu_0, \frac{\sigma}{\sqrt{n}}) = \mu_0 + z_{\frac{1}{2}\alpha} \cdot \frac{\sigma}{\sqrt{n}}$$

$$C_L = \text{invNorm}(\frac{1}{2}\alpha, \mu_0, \frac{\sigma}{\sqrt{n}}) = \mu_0 - z_{\frac{1}{2}\alpha} \cdot \frac{\sigma}{\sqrt{n}}$$

So

$$\beta = P(H_0 \mid H_1) = P(C_L \leq \bar{X} \leq C_R \mid \mu = \mu_0 + \delta)$$

$$= P(\bar{X} \leq C_R) - P(\bar{X} < C_L)$$

$$= \Phi\left(\frac{\mu_0 + z_{\frac{1}{2}\alpha} \cdot \frac{\sigma}{\sqrt{n}} - (\mu_0 + \delta)}{\frac{\sigma}{\sqrt{n}}}\right) - 0$$

See Fig. 9-1 (tutorial)

$$= \Phi\left(z_{\frac{1}{2}\alpha} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

So

$$z_\beta = -z_{\frac{1}{2}\alpha} + \frac{\delta\sqrt{n}}{\sigma}$$

Therefore

$$n \geq \frac{(z_{\frac{1}{2}\alpha} + z_\beta)^2 \cdot \sigma^2}{\delta^2}$$

Example 3

We look again at: $H_0: \mu = 300$ against $H_1: \mu \neq 300$

$$\alpha = 0.05$$

$$X \sim N(\mu = ?; \sigma = 15)$$

How large should n be chosen so that the test with a probability of at least 90% detects a difference of size 8 compared to H_0 ? In other words β must be less than or equal to 0.1.

Solution

$$\left. \begin{array}{ll} \mu = 308 & \Rightarrow \delta = 8 \\ \sigma = 15 & \end{array} \right\} \Rightarrow n \geq 36.9402$$

$$\left. \begin{array}{ll} \beta = 0.1 & \Rightarrow z_\beta = 1.2816 \\ \alpha = 0.05 & \Rightarrow z_{\frac{1}{2}\alpha} = 1.96 \end{array} \right\}$$

So choose $n = 37$.